

Relative size of brain and brain components in three gliding placentals (Dermoptera: Rodentia)

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The relative weight and the volumetric composition of the brains of *Cynocephalus*, *Iomys*, *Glaucomys*, and pteropodids are compared. It appears that in three well separated mammalian orders and at different places and times, a fairly similar subdivision of the brain was brought about by evolution, which is considered to be a result of a general parallelism in ecological conditions and behaviour.

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On compare le poids relatif et la composition volumétrique des cerveaux de *Cynocephalus*, *Iomys*, *Glaucomys* et de pteropodidés. On montre que, dans trois ordres bien séparés de mammifères et à des endroits et époques différents, l'évolution a produit une subdivision assez semblable du cerveau, ce que l'on considère comme résultant d'un parallélisme général de conditions écologiques et de comportement.

Introduction

Among the examples of evolutionary convergence in placental mammals, that of the "flying lemur" of Southeast Asia and the "flying squirrels" of Asia, Africa, and North America is most remarkable. In fact, one of those animals actually flies but all can glide considerable distances, for example up to 136 m for the flying lemur (Walker 1975) and not much less for some Asiatic sciurids according to information collected in Japan by Pirlot. They all support themselves through the aid of a broad fold of skin that extends between arm and leg on each side of the body. Outside the realm of placentals, a similar organ with a similar function is found among Australian marsupials (Phalangeridae) and this makes an even more striking convergence. We shall be concerned here with (1) the flying lemur *Cynocephalus* commonly called *colugo* or *cobego* in Southeast Asia (Anonymous 1959), the species in our material being presumably *C. variegatus*, a member of the family Cynocephalidae (Harrison 1966) of the order Dermoptera; (2) the flying squirrel *Iomys horsfieldii* or red-tailed flying squirrel of Malaysia; (3) the northern flying squirrel *Glaucomys sabrinus* known under the name of *polatouche* in the Province of Quebec, Canada. Both squirrels belong to the order Rodentia, suborder Sciuromorpha, family Sciuridae, subfamily Petauristinae.

The flying squirrels are mainly fruit eaters but they also take pieces of barks, buds, and fungi, and occasionally insects. The Dermoptera live on a diet of fruit,

flowers, buds, and in large amount, leaves (Walker 1975).

The evolutionary split between the Dermoptera and the Rodentia goes back a long way, at least to the early Cenozoic and possibly into the late Mesozoic. Simpson (1945) writes that the Dermoptera "are very aberrant derivatives of the Insectivores, of the insectivore ancestry of the Chiroptera, or of the earliest Chiroptera." He mentions the Upper Paleocene as the earliest time for known fossil Dermoptera, of the family Plagiomenidae (North America), the family Cynocephalidae being recent. According to the same author again, it seems that the oldest fossil rodents were also found in the Upper Paleocene and those are Sciuromorpha of the extinct family Ischyromyidae. The oldest fossil Sciuridae are present in Miocene deposits and the Petauristinae are known from the Pleistocene.

Anyone interested in morphofunctional adaptations is likely to ask whether the sensory-motor centers of those three convergent gliders show any common feature related to their particular locomotion. This question refers not only to the art of gliding or flying considered in itself, but also to other habits correlated with it. Being an airborne vertebrate is commonly believed to imply a particular organization of the brain (Jerison 1973, p. 157) and especially of the cerebellum (Romer 1962, p. 539). The parts played respectively by olfaction, vision, and audition may certainly differ in relation to whether an animal walks or swims or flies; consequently, the structure and (or) size of the corresponding brain areas

may differ in a parallel way. This is a general assumption in classical neurology and one that previous work by Pirlot and co-authors (Pirlot 1970, 1974, 1977, 1980, 1981; Pirlot and Pottier 1977; Pirlot and Stephan 1970; Stephan and Pirlot 1970; Stephan et al. 1974) has substantiated. To elucidate those questions in the case of our gliders, one may start by looking at the brain of each species, in order to see if its general organization bears any similarity in all three types. We have tried to do that by carrying out a simple volumetric study of the brain-body relationship in those three mammals.

Like previous papers on bats, edentates, and marsupials (Pirlot 1980, 1981; Pirlot and Pottier 1977; Pirlot and Stephan 1970), the present article will attempt to show that certain associations may exist between a particular quantitative composition of the brain and particular life habits of the species concerned. That means we will try to detect a special relation between brain morphology and behaviour, which in turn means a definite relation to ecological conditions. This work thus belongs in functional morphology since it aims at characterizing ecological types through significant morphological features.

Materials and methods

Materials

A female *Cynocephalus* was obtained by Kamiya from contacts in Bangkok, Thailand; its body weight was 810 g and its brain weight 6.2 g. A female *Iomys* was caught by Pirlot near a Dayak village in the jungle of Borneo; its body weight was 119 g and its brain weight 2.2 g. A male *Glaucornis* was collected by students on the grounds of the Station de Biologie de l'Université de Montréal, 70 km northwest of Montreal, Canada; its body weight was 49.5 g and its brain weight 0.938 g; this was obviously an immature individual with very small testes (the adult weight varies from 75 to 125 g). All three animals were perfused intracardially with 10% formalin. The brains were embedded in paraffin, cut at 10 µm, and Nissl stained in Pirlot's laboratory in Canada. The study was completed jointly by both authors in Kamiya's department in Japan.

Methods

The approach is quantitative and follows the procedure successfully initiated by Stephan (1967) then applied by him and other authors to insectivores, primates, Chiroptera, Edentates, etc. That method can be characterized as "basic" in the sense that it is hardly more than introductory, as a biometrical treatment, but at the same time indispensable before any deeper quantitative analysis of brain structures is attempted. Its successive steps have been explained elsewhere on several occasions, for example in Pirlot and Stephan (1970) and Stephan et al. (1970). The principle of the allometric comparisons is simple: (1) the classical allometry equation $y = bx^k$ relating brain size (y) to body size (x) is applied in its log form $Y = B + kX$, where $Y = \log y$, $B = \log b$, and $X = \log x$, while k is the growth exponent; (2) a composite group called the "basal Insectivores" by Stephan (1967) is taken as reference, i.e., the constants of the equation $Y = B + kX$ for that group are used to estimate the brain sizes of basal

insectivores whose body sizes would be equal to those of the animals under investigation; (3) the ratios of the actual brain sizes in the animals studied to the expected brain sizes in the corresponding basal insectivores are taken as progression indicators, i.e., they express how many times the brains under investigation are larger than the brains of those hypothetical insectivores of same body size; (4) the same procedure is applied to the main brain components as delimited on photographs of brain sections. Multiplying the ratios (actual/expected) by 100, one obtains progression indices which may vary from below 100 to above 100, the 100 level indicating the theoretical expected value in the reference insectivores. Body size and brain size were expressed by weight and brain-component size by volume as in previous studies.

One brain only was studied from each species reported upon here. It must be realized that, in practice, such a work cannot be possibly carried out in a statistical fashion, nor does it need to be so. It involves the lengthy task of embedding and sectioning the brains, then photographing a number of sections (about 125 for each brain); after that, the various components must be delimited and clipped or traced with the stylus of an image analyzer. These operations provide the raw figures from which the volumes can finally be estimated (for technical details, see Stephan et al. 1970). In many cases, in particular with the present material, collecting large samples of specimens would take years and require more money than most researchers can afford; in the case of *Cynocephalus*, it simply is out of the question for reasons of nature conservation. Furthermore, in the few cases (see Stephan and Pirlot 1970; Pirlot and Pottier 1977) where two or three brains from the same species were available and sectioned, it appeared that the interindividual variability was so low as not to confuse an estimation of the intergroup differences (intergeneric, inter-familial levels). Other questions pertaining to our methodological approach may be asked. We shall discuss some of them in our comments further below.

Results

The quantitative results for total brains and for brain components will be considered separately. For the sake of comparison, we will also refer to data previously published on frugivorous and nectarivorous bats.

Total brains

Applying the method described above to the brain weight (BrW)/body weight (BoW) relationship, one obtains the following progression indices: 213 for *Cynocephalus*, 253 for *Iomys*, and 187 for *Glaucornis*. In three pteropodid bats (frugivorous Megachiroptera), the indices are 235 for *Cynopterus horsfieldii* (with BoW 53 g), 266 for *Rousettus aegyptiacus* (with BoW 148 g), and 271 for *Pteropus vampyrus* (with BoW 1220 g) (BoW from Stephan et al. 1974). We have selected here three Chiroptera with BoW more or less comparable to those of our Dermoptera and Rodentia. It can be seen that there are no large differences between all the indices except in the case of *Glaucornis* which exhibits a rather low encephalization although it is an immature individual (young mammals tend to have relatively high BrW/BoW ratios but the progression

concept refers to the BrW/BoW of a basal standard type with which a comparison is made, as explained above).

Concerning our comparisons with bats, it must be noted that the sample borrowed from Stephan et al. (1974) displays a much broader size range than our gliders. In several personal discussions, the progression index method, as applied to such materials, has been considered with suspicion by some biometricians while readily accepted by others. A debate on this problem is outside the scope of the present paper. Excessive reliance on linearity or on the evolutionary significance of the indices may have occurred: only big differences in development can be detected with indices. It was suggested that the indices may tend to increase with absolute body size, without any relation to a functional progress involved in the structures. It is possible that this has occurred with certain materials, perhaps in taxonomically heterogeneous and small samples; but it is not observed in the homogeneous group of pteropodids referred to here. The reader can verify this for himself by looking at Tables II and VIII in Stephan et al. (1974) and considering all seven species together, as well as Table 5 in Stephan and Pirlot (1970) and Table III in Pirlot and Pottier (1977), again considering all the species. We give in Tables 1 to 3 the ranges of variations for the pteropodids: whatever values are selected from within these ranges will not change our conclusions hereafter.

Brain components

The simple ratios of 11 brain components to total brain are shown in Table 1. The ratios of some telencephalic components to total telencephalon are presented in Table 2. The progression indices for the same brain components are given in Table 3. The pteropodids appearing in those tables do not include *P. vampyrus*, whose brain was not available in sections, but a smaller species, *P. lylei* (BoW = 314 g).

Comments

Encephalization

It is not always easy to gauge the significance of equality or inequality between progression indices for total brain. Here, we are dealing with two ancient and distinct orders. One, the Rodentia, has proved successful over 70 million years, mainly thanks to the ecological versatility of its members. Their sciurumorph branch, however, comprises some families that are less "generalized" than those found, for example, among the Myomorpha. Furthermore, among the Sciuromorpha the gliders probably are the most specialized morphologically. In its life habits, *Cynocephalus* is close to the flying squirrels, that is in its locomotion and vegetarian diet. The reason why *Glaucmys* has a lower encephalization index is unknown. All indices (187 to 253) are also close to those obtained for seven pteropodid bats (range 199 to 272 for six of these, the seventh one reaching, exceptionally, 382 (Stephan et al. 1974)).

This comparison with the pteropodids is made here on rather general grounds, i.e., the pteropodids abound in the same area as two of our gliders; they originated early in the Cenozoic as did at least some of the gliders; they are strictly arboreal and frugivorous, again like the gliders. Of course, in contrast to our gliders, they are true flyers; we do not know yet to what extent this characteristic may entail differences in the adaptation to the environment as far as the overall brain structure is concerned. The question of the exact significance of the encephalization progression indices must be left open for the time being. It would be interesting to see if the indices obtained for our gliders contrast significantly with those estimated for a spectrum of species with other (i.e., nonvegetarian) diets and other niches in general. Too little work has been done so far with the same method to permit far-reaching conclusions. We can only refer to limited observations. The insect-eating Microchiroptera studied by Pirlot and Stephan (1970) and Pirlot and Pottier (1977) have clearly lower encephalization indices than our vegetarian gliders and all vegetarian bats investigated (usually well below 170). On the other hand, data supplied by Pirlot (1980) show that two edentates, one insectivorous and one vegetarian, have the same low index of encephalization (about 146). These examples point to the limitation of the indicative value of the encephalization index with respect to diet. More data on relative brain development in mammals of varied sizes and food preferences will have to be gathered before one can appraise exactly the ecological significance of that index. In contrast, as we shall see below, some brain-component indices provide clearer criteria.

Brain composition

We will comment on selected features of the brain composition only.

Neocortex and schizocortex

Before discussing the volumetric data, we must have a look at the gross external morphology of the neocortex in the brains under study (Fig. 1). It is striking that *Cynocephalus* possesses a neocortex with well marked sulci. On its dorsal surface, there is a clear sulcus lateralis. In lateral view, we see a deep sulcus rhinalis with a correspondingly well developed pyriform lobe. The sulcus rhinalis branches cephalad, giving off a sulcus orbitalis (or supraorbitalis) while it continues caudad and dorsad as a sulcus pseudosylvius homologous to a sulcus sylvius. The posterior ventrolateral angle of the neocortex is marked off as a gyrus postsylvius (rather than a gyrus hippocampi as in Grassé, 1955, apparently from an earlier inaccurate description by Leche). In contrast to the colugo, both squirrels exhibit lissencephalic hemispheres with only a shallow sulcus rhinalis. A faint groove on the dorsal surface of the neocortex in *Iomys* may be the first sketch

TABLE 1. Brain components in percentage of total brain and volumes (absolute volumes in parentheses)

Brain component*	<i>Cynocephalus</i>	<i>Iomys</i>	<i>Glaucomys</i>	Pteropodids†
BO	2.25 (130.5)	2.08 (43.5)	2.70 (24.2)	4.15 (31-206)
N	37.65 (2175.2)	33.68 (705.3)	35.10 (315.8)	34.70 (223-2385)
RH	10.33 (599.0)	6.44 (135.0)	7.97 (71.7)	7.18 (64-376)
S	0.77 (44.5)	1.11 (23.2)	0.95 (8.5)	1.48 (15-69)
D	11.17 (645.5)	10.59 (221.9)	11.53 (103.7)	9.45 (79-514)
St	4.20 (242.4)	6.13 (128.4)	5.51 (49.5)	6.13 (38-392)
H	5.79 (335.1)	7.88 (165.1)	7.57 (68.0)	6.72 (81-287)
M	6.19 (357.8)	6.96 (145.7)	4.99 (44.9)	5.43 (47-238)
C	11.02 (636.8)	13.70 (287.1)	11.71 (105.4)	13.60 (117-710)
O	7.21 (416.5)	9.79 (205.1)	9.42 (84.7)	7.60 (78-325)
Sz	3.42 (198.0)	1.64 (34.2)	2.55 (22.9)	3.56 (25-186)
%	100.00	100.00	100.00	100.00
Total	(5781.1)	(2094.6)	(899.3)	(822-5701)

*BO, bulbus olfactorius; N, neocortex; RH, rhinencephalon; S, septum; D, diencephalon; St, striatum; H, hippocampus; M, mesencephalon; C, cerebellum; O, medulla oblongata; Sz, schizocortex.

†Average for seven species estimated from Stephan et al. (1974).

TABLE 2. Percentage data on the telencephalon*

	Tel/Br	N/Tel	RH/Tel	St/Tel	H/Tel
<i>Cynocephalus</i>	64.07	62.49	17.93	6.55	8.30
<i>Iomys</i>	58.96	57.12	10.92	10.39	13.36
<i>Glaucomys</i>	62.35	58.72	12.78	8.83	12.14
Pteropodids†	61.98 (59-68)	50.03 (44-61)	12.22 (9-14)	9.36 (7-11)	12.93 (7-16)

*Tel, telencephalon; Br, total brain; N, neocortex; RH, rhinencephalon; St, striatum; H, hippocampus.

†Average for seven species from Stephan et al. (1974).

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of a sulcus lateralis. By classical standards, *Cynocephalus* appears to be significantly more advanced in cortical differentiation than either *Iomys* or *Glaucomys*. On the other hand, the brains of large Megachiroptera exhibit some of the surface features of that of *Cynocephalus* and they differ from the brains of Microchiroptera in that respect.

Another observation of macroscopical anatomy is of interest here. In side view, *Cynocephalus* shows a

well-shaped, clearly convex caudal portion of the neocortex, contrasting with the blunt (*Iomys*) or concave (*Glaucomys*) outline of the same area in the squirrels. That part of the brain includes in particular the visual cortex. *Cynocephalus* is said to be nocturnal (Grassé 1955) or dusk active (Walker 1975; Wharton 1950). The caudal brain of *Cynocephalus* resembles in shape that of several pteropodid bats. The latter, as observed by Pirlot in Africa and Asia, use their eyesight a great deal before

TABLE 3. Progression indices for brain components

Brain components*	<i>Cynocephalus</i>	<i>Iomys</i>	<i>Glaucomys</i>	Pteropodids†
BO	44	51	49	104 (69–144)
N	582	657	520	621 (471–1182)
RH	129	92	83	105 (76–143)
S	105	167	102	233 (184–285)
D	315	356	286	323 (244–452)
St	203	347	229	333 (213–588)
H	165	242	165	229 (174–266)
M	250	304	154	247 (220–286)
C	171	284	189	285 (234–364)
O	126	186	126	152 (140–169)
Sz	302	156	173	294 (232–410)

*Abbreviations as in Table 1.

†Average for seven species estimated from Stephan et al. (1974).

at sunset and before sunrise, in contrast to the mainly acoustic and truly nocturnal Microchiroptera. Furthermore, in pictures published by Morris (1965, p. 89) and Walker (1975, p. 181), one may notice that *Cynocephalus* possesses large eyes, as in the pteropodids. The problem of the relative development of the visual cortex in subnocturnal wild mammals is well worth more biological investigation. The contrast with the two squirrels is puzzling: the latter are diurnal species with presumably good eyesight and well organized visual cortex.

Quantitatively, the neocortex is the largest brain component and it is, in our three animals, as in many mammals, the most progressive of all. As expected from the encephalization index, *Glaucomys* stands lower than the other two genera in cortical progression. But all three fall within the range observed in the seven pteropodids (471 to 636 plus the exceptional value 1182 for one large genus (Stephan et al. 1974)). The high percentages taken by the telencephalon within the brain and the neocortex within the telencephalon in *Cynocephalus* are also remarkable (Table 2). However, the neocortical index for *Cynocephalus* (582) is somewhat lower than that for *Iomys* (657) but there is here an uncertainty due to the difficulty in delimiting the schizocortex. Indeed, it is quite possible that the area identified as schizocortex in the laterocaudal portion of the brain is in reality smaller, so that the index 302 for schizocortex is a little too high while the figure 582 for the neocortex is too low, although the difference cannot be great (maybe 10 to 20 points). This correction puts neocortex and schizocortex at practically the same level in *Cynocephalus* as in the pteropodid bats. On the other hand, *Iomys* (657) unquestionably possesses a more progressive neocortex than *Glaucomys* (520).

One must dispel here the belief, implied among others by Grassé (1955) that *Cynocephalus* is a lowly corticalized species because its hemispheres do not cover its

mesencephalon completely (colliculi posteriores). Indeed, *Cynocephalus* seems to have a shorter neocortex than either of the two squirrels (Fig. 1), but that superficial observation is misleading. To appreciate the real relative size of the neocortex, volumetric measurements alone are reliable. In that respect Tables 1 and 3 demonstrate that the neocortex in *Cynocephalus* is by no means underdeveloped.

Finally, we wish to draw attention to the very high indices for the schizocortex in *Cynocephalus* (302, with the restriction expressed above) and the bats (294), in contrast to the only moderately large indices (156 and 173) in the two rodents.

Bulbus olfactorius and rhinencephalon

The three gliders show a strong regression of the bulbus (44 to 51) contrasting with the nonregressive average figure (104) for the pteropodids. Note that the latter are exceptional among bats, all other Chiroptera being regressive in that respect (Stephan and Pirlot 1970; Pirlot and Pottier 1977). But while the rhinencephalon is accordingly little developed too in the squirrels (92 and 83), it is relatively high in *Cynocephalus* (129). In this animal, there is thus a discrepancy between the two olfactory components, bulbus and rhinencephalon. We believe the reason for this is that we delimited the rhinencephalon from the sulcus rhinalis, as is usual. In the case of *Cynocephalus*, there is a very large pyriform cortex that falls into the rhinencephalon when that practice is followed but in the two squirrels the typical pyriform cortex is relatively much smaller. This points to (a) the probably nonolfactory significance of the pyriform area; (b) the great size variability of this area, which raises the problem of its functional significance between and within mammalian orders. It is logical to expect a correlation between the bulbus and the rhinencephalon. But one should be aware that the rhinencephalon, as structurally defined in traditional

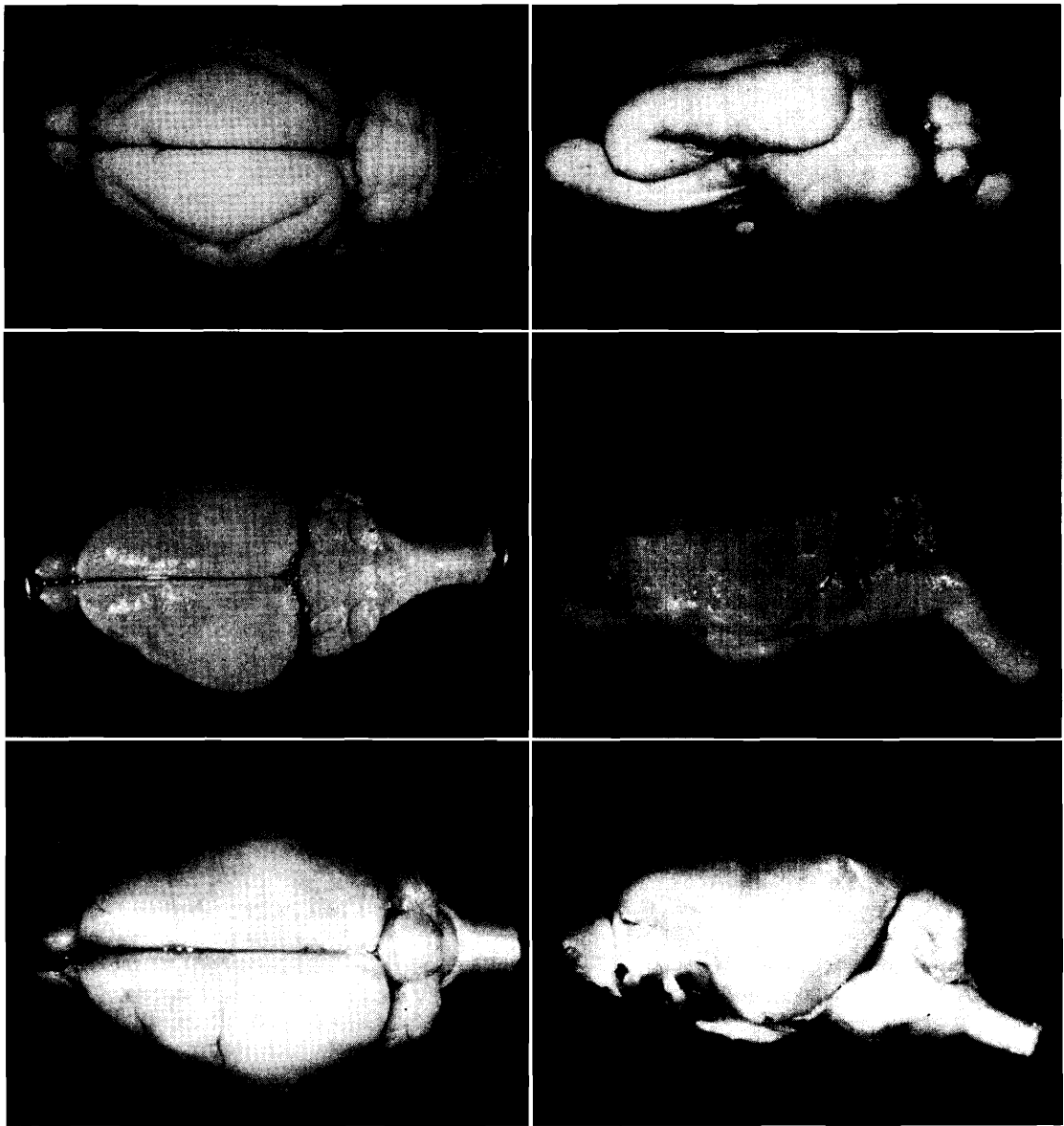


FIG. 1. External aspects of the brains. Left column, dorsal views. Right column, lateral views. Top, *Cynocephalus*. Middle, *Iomys*. Bottom, *Glaucomys*. Specimens reduced to same size on photographs. Actual lengths from bulbi to cerebellum after perfusion with formalin: *Cynocephalus*, 38 mm; *Iomys*, 26 mm; *Glaucomys*, 19 mm.

anatomy, i.e., ventral to the sulcus rhinalis, may include regions that have diverged *functionally* in the course of evolution between orders or families or genera. To assume a strict quantitative parallelism between the bulbus and the rhinencephalon in all cases may be unjustified, because such a correlation simply may not exist in many instances. In that respect, a statement by Stephan et al. (1974, p. 178) to the effect that "the variations of (the rhinencephalon) are in agreement with those of the olfactory bulb" was rather premature in view of the fact that Table VIII, p. 177, of the same paper

shows that, in seven Megachiroptera, the bulbus is more progressive than the rhinencephalon in three cases and less progressive in four cases. The same generalization was repeated by Pirlot and Pottier (1977) following Stephan and Pirlot (1970). In fact, it is probably more valid for Microchiroptera. Obviously, the morphofunctional relations between the bulbus and the rhinencephalon deserve a second look in possibly heterogeneous orders such as the Chiroptera and the Rodentia; the first thing to do in that respect is to distinguish histologically the rhinencephalon proper, the amygdaloid region, and

above all the pyriform cortex, as was attempted in a preliminary way by Pirlot (1980) in the case of two edentates.

As far as *Cynocephalus* is concerned, referring to this animal as macrosomatic (Grassé (1955), probably after Leche again) does not seem to be warranted by our quantitative analysis. The most significant component in that respect is the bulbus, not the rhinencephalon, especially not a rhinencephalon including a large pyriform area. The bulbus alone, probably, provides a reliable indication of the intensity of the olfactory activity. It must really be looked upon as almost peripheral, at most as a first sensory relay. In mammals, it has generally though not necessarily regressed from the ancestral insectivore condition. In *Cynocephalus* it is strongly regressive just as in the squirrels, and this contrasts with its development in some pteropodids (the numbers for the latter vary from 69 to 144 (Stephan et al. 1974)).

Cerebellum

A surprising difference exists between *Iomys* (284) on the one hand and *Cynocephalus* (171) and *Glaucomys* (89) on the other hand, in the development of the cerebellum. Further, *Iomys* (284) is equal to the pteropodids (285) in that respect. As already pointed out in the Introduction, one would expect true flyers like bats and also good gliders to have a well developed cerebellum. This, at least, is again a classical assumption; Stephan and Pirlot (1970) have shown that it may lead to an oversimplification of reality and further data from Pirlot and Pottier (1977) and Pirlot (1980) suggest that flight is not the behaviour that demands the most sophisticated cerebellar control: for example, precise and delicate movements such as those of a vampire climbing on a sleeping person, incising his earlobe, lapping the blood, and retreating without awakening the sleeper (Pirlot 1965; Stephan and Pirlot 1970) is a much more difficult venture, and so is the handling of objects with fingers or claws as occurs in primates and even more "modest" mammals with relatively well developed cerebellum (e.g. edentates (Pirlot 1980)). However, we cannot elaborate further on the cerebellar indices obtained for our gliders in the absence of details on their behaviour.

Other components

In a general way, *Iomys* shows much more progression than the other two mammals and it is closer to the pteropodid bats (Table 3). For most components, *Cynocephalus* and *Glaucomys* keep close to one another.

Conclusion

The brains of the three gliders which we have examined with a quantitative method exhibit certain common features and they also share some characteristics with pteropodid bats. This general result is not unexpected in view of the fact that (a) these mammals

display certain similarities in their behaviour, being arboreal, vegetarian, and (in different ways) airborne animals; (b) the Dermoptera and the Megachiroptera probably acquired their main group characteristics at about the same time in Cenozoic evolution, originating presumably from early insectivores that were not yet greatly differentiated; the same can be said of the Sciuromorpha to which our two Petauristinae belong, although the latter appeared much later. This parallelism is of course limited and must not be overemphasized, but the important fact is that it does exist. Brain structure seems, in this case, to be related at least in some measure to partly common life habits, and thus this is probably a good example of sensory ecoethology, a fashionable but not easily documented approach. This question has been discussed before (Pirlot and Stephan 1970; Pirlot 1974).

Do our results shed any new light upon the present distribution of the three gliders? This is another problem. Adaptation to gliding in forests apparently arose independently in various places and times. In all instances, the brain was affected partly in similar, partly in dissimilar, manners. In the case of the early Cenozoic Dermoptera, the evolution also involved deep transformations of other morphological features (limbs, dentition, etc.). Something analogous, with greater specialization in locomotion (true flight), occurred at about the same time in Megachiroptera. Were the first Dermoptera already gliding animals? That question is difficult to answer; but, considering the peculiar morphological characteristics of that order and its similarity with the Megachiroptera in brain organization, it seems very likely that it acquired the gliding ability from the beginning of its history. In passing, it must be noted here that the pteropodids are the most, perhaps the only, significant gliders among bats (Pirlot 1977; Pennycuik 1973). The flying sciurids took essentially the same course although possibly much later: they may be younger and are still members in good standing of the order Rodentia, not having diverged sufficiently to constitute their own order; some gliding squirrels, however, belong to another family (the Anomaluridae), and live isolated in Africa; the Anomaluridae are known with certainty from the African Recent but some uncertain Anomaluroidea have been reported from the European Upper Eocene (Simpson 1945). It appears that there have been repeated evolutionary opportunities for small mammals to take up the gliding frugivorous life habits in boreal, temperate, and tropical forests. One species filled that niche among the Canadian boreal trees, others in the temperate woodlands of North America and Japan, still others in the tropical jungles of Asia and Africa south of the Sahara. Taxonomically separate but ethologically convergent gliding marsupials (Phalangerids (Pirlot 1981)) achieved a like specialization in Australia. A certain degree of convergence between the brains of all those mammals should not

be more surprising than the one that is said to exist between the brains of flying reptiles and those of birds (Jerison 1973). This author also writes: "The brain of flying mammals is involved in adjustments to aerodynamics and in the coordination of powered and gliding flight ..." (Jerison 1973, p. 157). We would agree with that statement although it is only part of the story.

In the above perspective, one may wonder in what sense exactly the expression "very aberrant derivatives" (Simpson 1945) can be applied to the Dermoptera. The colugo does show some peculiar morphological traits that make it look aberrant. But no systematist denies its relationship with an ancient insectivore stock which confers to it a certain *conformity* with other derivatives from the same insectivores. One may be tempted to regard it as a relict animal, hence as an archaic one, because of its relative taxonomic and geographic isolation. The ideas of "aberrance" and "archaism," however, should be tempered with the observation that, in its brain structure, *Cynocephalus* conforms well with the pattern of one of our widely spread and recent ecoethological types. It is simply an Asiatic local form which "decided" to become a glider, perhaps very early in the Cenozoic, and whose quantitative brain organization is "retarded" but level with that of other gliding insectivores (especially the American genus); in fact, it is even ahead of them all in some respects. If such is the case, and taking our new observations into account, we cannot accept without reservation the statement by Wharton (1950, p. 270) that "Lemurs are known to be of a low order of intelligence." That author seems to derive his opinion from watching a captive and panicky *Cynocephalus* jump on anything within reach and "start up a post supporting a house as if it were a tree" (Wharton 1950). What does that "mistake" really prove? With which other species should one compare *Cynocephalus* in intelligence? Not with monkeys or apes, to be sure.

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